

SMART LUMINESCENT COATING FOR IMPACT DAMAGE DETECTION ON COMPOSITE LAMINATES

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Abstract

Composite materials are widely used in aerospace structures due to their high strength-to-weight ratio, but they remain vulnerable to impact. Conventional structural health monitoring methods, based on embedded or surface-mounted sensors such as fiber Bragg gratings or piezoceramics, can compromise laminate integrity and introduce issues related to wiring, connectors, and electromagnetic interference. These limitations increase system complexity, weight and cost, while reducing reliability.

Mechanoluminochromic (MLC) materials offer an alternative by exhibiting fluorescence changes under mechanical stress, although their limited robustness hinders direct use. This work presents a smart coating that incorporates a copper-based hybrid coordination polymer into a thermoset matrix, combining MLC sensitivity with the mechanical durability and processability of paints.

The multifunctional coating was spray-deposited onto a cross-ply woven carbon fiber reinforced plastic laminate and tested under low velocity impact at varying energy levels, inducing barely visible damage. Under UV light, impacted regions were imaged using a digital camera, and fluorescence intensity was quantified via a green-to-blue ratio using a custom MATLAB algorithm. The optical response showed strong correlation with absorbed impact energy and damage, validated through micrographic cross-section analysis.

The MLC coating enables full-field, non-contact impact damage detection, supporting its potential as a next-generation non-destructive inspection (NDI) technique for advanced composite applications.

Keywords: *Mechanoluminochromic, smart coating, impact detection, composite laminate, NDT*

1 Introduction

Composite materials are extensively utilized in aerospace structures due to their high strength-to-weight ratio and mechanical performance. Nevertheless, their vulnerability to impact events, often resulting in barely visible damage (BVD), presents critical challenges for structural integrity and operational safety. Structural Health Monitoring (SHM) systems have been developed to address this issue, typically relying on embedded or surface-bonded sensors such as fiber Bragg gratings and piezoelectric transducers for impact detection and damage evaluation. Despite their effectiveness, these systems suffer from several drawbacks. Sensor fragility may adversely affect the mechanical performance of the host laminate, while the requirement for complex wiring or wireless systems introduces electromagnetic interference, additional weight, and manufacturing challenges [1], [2].

Mechanochromic materials offer a promising alternative by providing an optical response to mechanical stimuli. This property enables full-field damage detection across composite structures without compromising their integrity or requiring complex sensor integration [3], [4]. Within this class, MLC materials are particularly attractive, as they rely on variations in emission spectra rather than absorption. Consequently, their response is less sensitive to environmental factors such as surface contamination or ambient lighting conditions [5].

However, the direct use of MLC compounds is limited by their poor mechanical properties. To overcome this issue, they can be incorporated into polymer matrices using two main strategies: intrinsically luminochromic polymers or additive-based systems. The former approach involves covalent integration of chromophoric units into polymer chains, resulting in materials with inherent MLC behavior. This method requires specific functional groups and complex synthesis routes and typically activates only under high loads with reversible optical responses [5], [6]. In contrast, the additive approach disperses MLC compounds within an inert polymer matrix, forming a biphasic system. This strategy offers greater flexibility in material design and simpler processing, avoiding complex chemistry. However, weak interfacial interactions can lead to poor dispersion and inefficient load transfer, again limiting activation to higher stress levels.

Alternative solutions, including piezochromic pigments (e.g., CuMoO_4) and mechanochromic polymers activated by bond rupture, have also been investigated [7], [8]. These systems have been applied as coatings or integrated into composite matrices for damage sensing, but challenges remain in terms of activation thresholds, signal stability, and fluorescence intensity. To address these limitations, this study investigates multifunctional coating based on MLC Cu-based hybrid coordination polymer (HCP), characterized by low activation threshold and tunable optical properties, dispersed within a thermoset epoxy matrix. The coating was applied to a cross-ply carbon fiber reinforced plastic (CFRP) laminate and evaluated under low-velocity impact (LVI) conditions. Fluorescence emission changes were correlated with impact forces using custom image-processing algorithms. The preliminary results demonstrate the potential of this MLC coating as a cost-effective, reliable, and easily implementable nondestructive inspection (NDI) technique for advanced composite structures.

2 Materials and Method

2.1 MLC coating formulation

The mechanoluminochromic additive used in this study is a copper iodide-based Hybrid Coordination Polymer, $[(\text{CuI})_3(\text{Br-py})]_n$, first synthesized by S. Masahara et al. [9]. It was selected for its strong UV-induced spectral response and white appearance under ambient light, making it suitable for smart coating applications. The material was produced as a polycrystalline powder via a slurry method by the Molecular crystal engineering group at the University of Bologna.

As reported by Contini et al. [6], mechanical grinding induces a shift in the emission peak of the MLC powder from blue (474 nm) to green (524 nm) under UV excitation.

The chromophores were dispersed in a two-component, water-based epoxy paint (Renner Italia). The MLC powder was mixed into the pre-polymer by stirring for 10 minutes before adding the hardener, with a pre-polymer/hardener ratio of 73:23 wt% and an MLC content of 3.5 wt% of the total formulation, following the procedure described in [10].

The coating was applied to composite panels by spray deposition using a conventional spray gun. Three layers were applied at 10-minute intervals to ensure uniform coverage, followed by curing at 60 °C for 48 hours to maximize the glass transition temperature (T_g).

In Figure 1 the differential scanning calorimetry (DSC) of the paint with and without MLC additive for 3 different curing cycles is reported. It can be observed that the presence of the additive minimally affects the maximum T_g of the epoxy matrix in which was dispersed: 32 °C for the

MLC coating vs 34 °C for the pristine one after 100 hours at 60 °C. Moreover, the complete curing of the MLC coating is reached after 48 hours at 60 °C: 33 °C after 48 h vs 34 °C after 100 h.

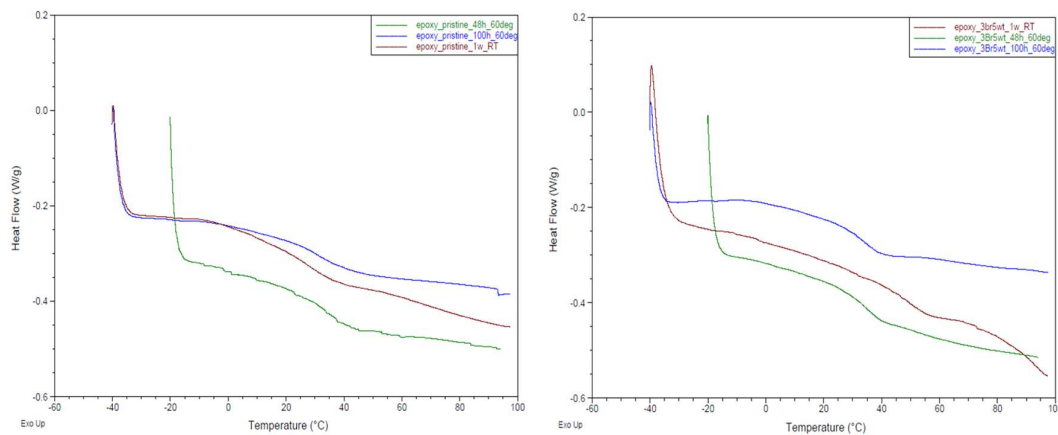


Figure 1 DSC scan of the water based epoxy paint, without (left graph) and with (right graph) MLC additive, after three different curing cycles: 48 hours @ 60 °C, 100 hours @ 60 °C and room temperature.

2.2 Composite substrate and LVI

The CFRP laminate used for the application of the MLC coating features a symmetric cross-ply stacking sequence $[(0^\circ/90^\circ)_2]_s$, manufactured from 300 g/m² T700 carbon fiber unidirectional (UD) epoxy prepreg with a nominal glass transition temperature (T_g) of 120 °C (DeltaPreg, UTS300 T700-DT120). The laminate was cured in an autoclave under vacuum and 6 bar pressure, with a 2-hour isothermal at 120 °C and heating and cooling rates of 2 °C/min. The final cured thickness was approximately 2.5 mm. Two panels were produced: one coated with the MLC paint (MLC) and one left uncoated (pristine). Specimens measuring 50 × 50 mm were extracted from both panels using diamond sawing.

Low-velocity impact tests were carried out on both laminate types to evaluate surface damage visibility in the uncoated specimens and the mechanocromic response of the MLC coating. Tests were conducted in accordance with ASTM D7136 [11] using a drop-weight impact tower equipped with a 1.3 kg impactor, a load cell, and a 12.7 mm diameter hemispherical steel tip. Due to the reduced specimen size, samples were supported on a plate with a central circular hole of 20 mm diameter. Impacts were performed at energy levels of 1.5, 3, and 6 J, with three specimens tested per condition. Following impact and optical characterization of the top surface, cross-sectional micrographic analysis of the laminates was conducted.

2.3 MLC emission analysis

The relationship between the absorbed impact energy and the emission spectral shift of the smart MLC coating was investigated through quantitative analysis of digital images acquired under ultraviolet illumination conditions. Impacted specimens were irradiated using a commercial UV torch emitting at a wavelength of 365 nm. All image acquisitions were performed in a darkened environment to minimize the influence of ambient light and improve signal-to-noise ratio in the recorded fluorescence.

Images were captured in RAW format using a 64 MP digital camera. To ensure repeatability and enable consistent comparison across all specimens and impact conditions, camera parameters were kept constant throughout the experimental campaign, according to the procedure described in [10]. Specifically, the aperture was set to f/1.79, the shutter speed to 1/15 s, and

the ISO sensitivity to 640. The equivalent focal length was fixed at 17 mm, while white balance was manually defined using coefficients (0.6045, 1, 0.6406) to preserve the spectral fidelity of the captured emission.

Following acquisition, the images were processed using a custom-developed MATLAB algorithm designed to extract quantitative information from the fluorescence response. The algorithm operates on a pixel-by-pixel basis, computing the ratio between the green and blue channel intensities (G/B). This ratio serves as an indirect metric of the emission spectral shift of the MLC additive, which is associated with mechanically induced structural changes in the crystal. Spatial mapping of the G/B ratio enables visualization of damage distribution, while averaged values within impacted regions were correlated with the absorbed impact energy, providing a quantitative link between mechanical loading and optical response.

3 Results

3.1 Impact Dynamic and Micrograph analysis

Impact damage on the composite panels was induced through low-velocity impact tests, following the procedure described in the previous section. Figure 2 presents the force–displacement curves obtained for the MLC-coated specimens at nominal impact energies of 1.5, 3, and 6 J. The dynamic response of the pristine specimens was found to be comparable.

At 1.5 J, the curve shows a negligible force drop, while the absorbed energy is significant if compared with the initial impact energy (0.85 ± 0.25 J vs. 1.5 J). This behaviour can be ascribed to dissipative phenomena or low-energy damage mechanisms, such as matrix cracking [10]. As the impact energy increases to 3 and 6 J, the dynamic response of the laminate shows a pronounced force plateau at approximately 2.5 kN, which is usually associated with fibre breakage and the consequent loss of bending stiffness. Moreover, the absorbed energy increases significantly, reaching 2.37 ± 0.26 J and 5.27 ± 0.26 J for 3 and 6 J, respectively.

In Figure 3, the cross-sectional micrograph of the laminate impacted at 1.5 J is shown. In agreement with the dynamic analysis discussed in the previous paragraph, transverse matrix cracks with a reversed pine-tree pattern can be observed (indicated by red arrows), suggesting the predominance of Hertzian contact stresses over bending effects. These small cracks can act as initiation sites for delamination under fatigue loading, which can subsequently propagate and lead to catastrophic failure of the composite structure.

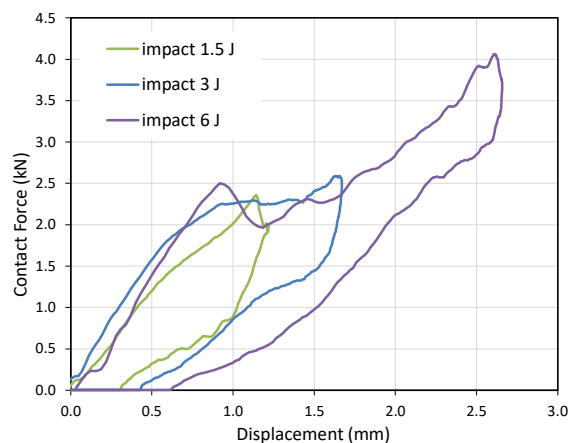


Figure 2 Low velocity impact test 1.5, 3, 6 J: Force vs displacement.

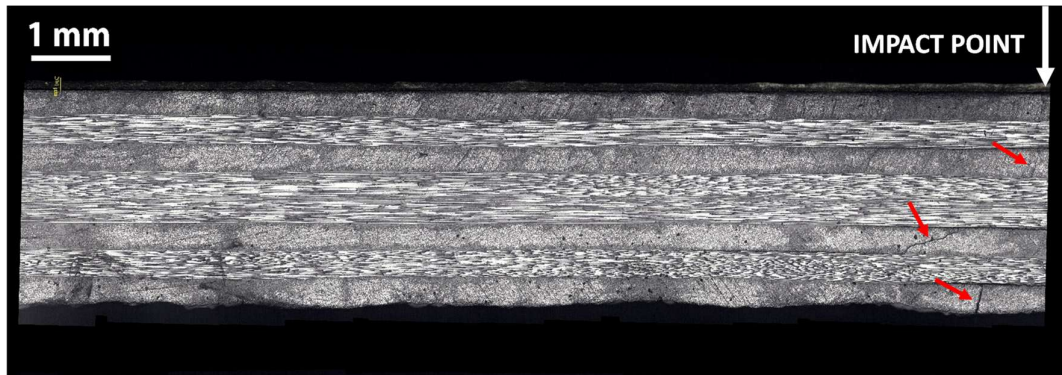


Figure 3 Cross-section micrograph of the MLC coated laminate impacted 1.5 J

3.2 MLC analysis

In Figure 4 image analysis of the 1.5 J impacted specimen top surface are shown under different conditions. The first two images from the left show the uncoated pristine specimen and the MLC-coated specimen illuminated by natural light. Both samples exhibit a slight indentation, along with whitening of the coating, which can be attributed to plasticization due to its relatively poor mechanical properties. The third image from the left shows the same MLC-coated specimen under UV illumination, in which the impact point becomes more clearly visible. Finally, the fourth image shows the G/B ratio of the third image, post-processed using the MATLAB algorithm described in the previous section. The G/B ratio exhibits a marked intensity increase in the impacted zone, due to damage of the MLC additive in the coating and the consequent shift in spectral emission from blue to green under UV illumination. These findings demonstrate the feasibility of the MLC coating as an impact detection and localization technology.

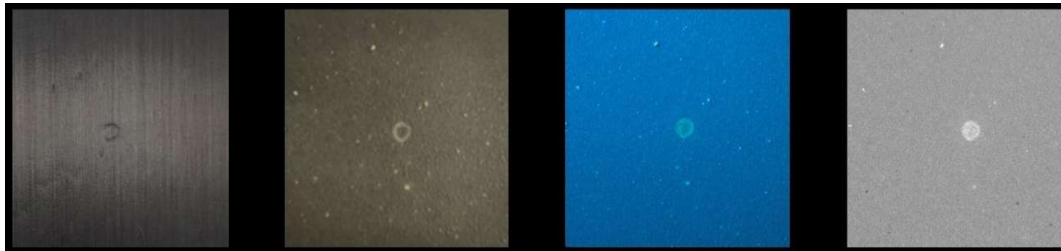


Figure 4 Image analysis of the top surface of the 1.5 J impacted specimens. From left to right: no coated and MLC coated under natural light, MLC coated under UV light, G/B ratio of the MLC coated under UV light.

The G/B ratio was integrated over the impacted zone area, for each impact energy and repetition. In Figure 5 the integrated G/B ration is plotted vs the absorbed impact energy. The quasi-linear correlation ($R^2 = 0.74$) between the two measurements highlights the potential application of the multifunctional coating not only for impact localization, but also for impact damage quantification.

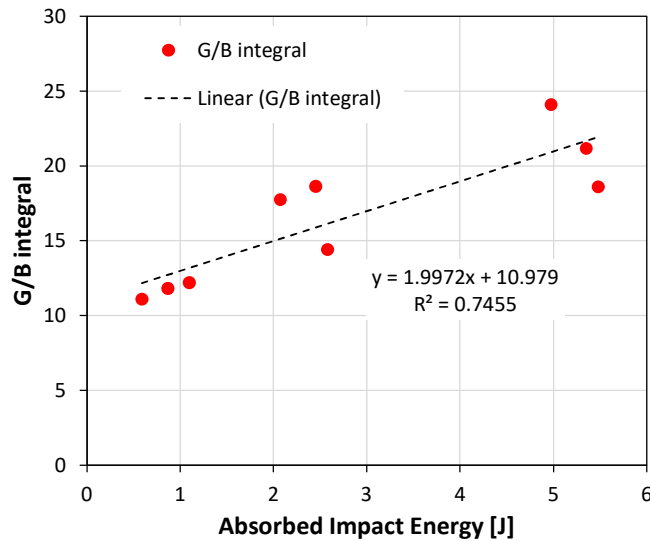


Figure 5 Integral of the G/B ratio on the impacted zone vs absorbed impact energy.

4 Conclusions

A sprayable mechanoluminescent (MLC) smart coating based on a Cu-based Hybrid Coordination Polymer was developed and validated for impact damage detection in CFRP laminates. The coating enables full-field, non-contact inspection without affecting the structural integrity of the substrate.

Under UV illumination, impacted areas are clearly identified through a fluorescence shift from blue to green, quantified via a G/B ratio of the digital image pixels. The optical response correlates with damage mechanisms observed in micrographs and with absorbed impact energy, showing a quasi-linear trend ($R^2 = 0.74$). This demonstrates the capability of the multifunctional coating not only for damage localization but also for preliminary damage quantification.

The proposed system represents a simple, cost-effective, and scalable solution for nondestructive inspection of composite structures, with strong potential for NDI applications.

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